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Operating Torque

Preliminary design calculations before friction tests could be preformed

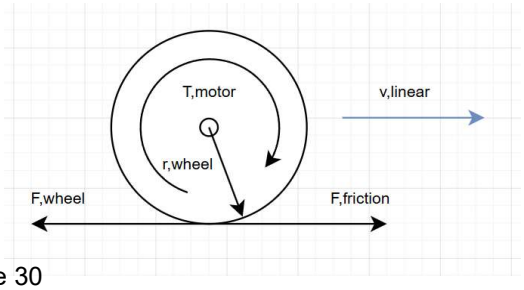
$T_{motor} := 0,5 \cdot 9,81 \text{ N cm} = 0,0491 \text{ J}$

$T_{both} := T_{motor} \cdot 2$

$m_{bot} := 500 \text{ g}$

$N_{bot} := m_{bot} \cdot g$

$g := 9,81 \frac{\text{m}}{\text{s s}}$



$T_{both} = 0,0981 \text{ J}$

$d_{wheel,outer} := 30 \text{ mm}$

$r_{wheel,outer} := \frac{d_{wheel,outer}}{2}$

$F_{wheels} := \frac{T_{both}}{r_{wheel,outer}}$

$c_f := 0,9$ est of polyurethane Shore 30 against an unknown surface

$a := \begin{bmatrix} F_{friction} \\ F_{wheels} \end{bmatrix}$

$F_{wheels} = 6,54 \text{ N}$

$F_{friction} := c_f \cdot N_{bot}$

$F_{friction} = 4,4145 \text{ N}$

$F_{push} := \min(a)$

$F_{push} = 4,4145 \text{ N}$

Stall Torque

$T_{stall} := 4 \text{ kgf cm}$ $F_{wheels} := T_{stall} \cdot \frac{2}{r_{wheel,outer}}$

$F_{wheels} = 52,3021 \text{ N}$

RPM needed

Theoretical / assuming no slip

$v_{robot} := 0,37 \frac{\text{m}}{\text{s}}$ Crossing roughly half of the arena

$rpm_{motor} := v_{robot} \cdot \frac{60}{\pi \cdot d_{wheel,outer}}$ $rpm_{motor} = 235,5493 \text{ Hz}$

Linear speed at the rated rpm

$rpm_{motor300} := 280 \text{ rpm}$

$v_{robot300} := rpm_{motor300} \cdot \pi \cdot \frac{d_{wheel,outer}}{60}$ $v_{robot300} = 0,0461 \frac{\text{m}}{\text{s}}$

Stall Safety Factor

N20 motors and Motor driver carrier

$I_{stall} := 1,2 \text{ A}$ $I_{md} := 1,7 \text{ A}$

$I_{SF} := \frac{I_{md}}{I_{stall}}$ $I_{SF} = 1,4167$

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Battery Voltage

$V_{max} := 12,6\ V$

$V_{min} := 9,3\ V \quad V_{nominal} := 11,1\ V$

Current and Discharge

$750 \cdot mAh \quad 45 \cdot C$ from listing

$I_{battery} := \left(750 \cdot (10)^{-3}\right) \cdot 45\ A \qquad I_{battery} = 33,75\ A$

$I_{motorstall} := 500\ mA$ Lower end estimate

$I_{MCU} := 25\ mA \qquad (max) \qquad I_{IR} := 40\ mA$

$I_{TOF} := 40\ mA$ Startup and shutdown peak $\qquad I_{IMU} := 3,6\ mA$

$I_{motordriver} := 3\ A$

$I_{system,peak} := 2 \cdot I_{motorstall} + I_{MCU} + 5 \cdot I_{TOF} + 3 \cdot I_{IR} + I_{IMU} + I_{motordriver}$

$I_{system,peak} = 4,3486\ A$

$SF_{current} := \frac{I_{battery}}{I_{system,peak}} \qquad SF_{current} = 7,7611$

Normal operation and run time est

$t_{run} := \frac{C_{3s}}{I_{max}} \qquad C_{3s} := 750\ mA\ hr \qquad I_{max} := I_{system,peak}$

$t_{run} = 620,8895\ s$

Thermal Calculations (heatsink)

$I_{5Vline} := I_{MCU} + 2 \cdot I_{IR} + I_{IMU} + 4 \cdot I_{TOF}$

Voltage measured at 11.49V after diode from LTSpice simulation

$I_{5Vline} = 0,2686\ A \qquad V_{afterdiode} := 11,49\ V$

$I_q := 0,005\ A$ Regulator datasheet (5 V)

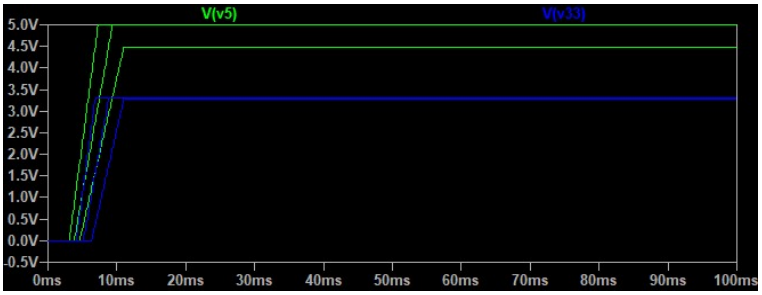
$V_{out} := 5\ V$

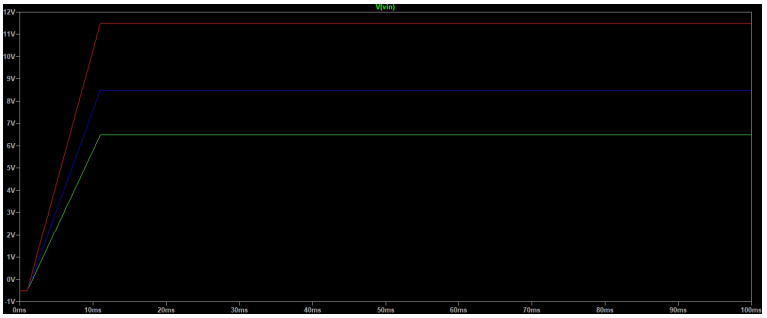
$P_D := \left(V_{afterdiode} - V_{out}\right) \cdot I_{5Vline} + V_{afterdiode} \cdot I_q$

$R_{CS} := 0,5\ \frac{°C}{W} \qquad R_{SA} := 50\ \frac{°C}{W} \qquad P_D = 1,8007\ W$

$R_{JC} := 5\ \frac{°C}{W}$

$T_{J1} := P_D \cdot \left(R_{JC} + R_{CS} + R_{SA}\right) \qquad T_{J1} = 99,9369\ °C$





Final junction temperature is around 110 deg at maximum operating conditions (TJ1+TA). 5 V regulator's maximum rating is 125

$T_{SF} := \frac{99,93}{125} \qquad T_{SF} = 0,7994$

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Real friction coefficient

$\theta_{shore30} := 68\ \text{deg}$ $m_{shore} := 10\ \text{g}$

$\theta_{shore40} := 64\ \text{deg}$ $m_{sebs} := 6\ \text{g}$

$\theta_{sebs} := 15\ \text{deg}$

$f_{s30} := m_{shore} \cdot g \cdot \sin\left(\theta_{shore30}\right)$

$f_{s40} := m_{shore} \cdot g \cdot \sin\left(\theta_{shore40}\right)$

$f_{ssebs} := m_{sebs} \cdot g \cdot \sin\left(\theta_{sebs}\right)$

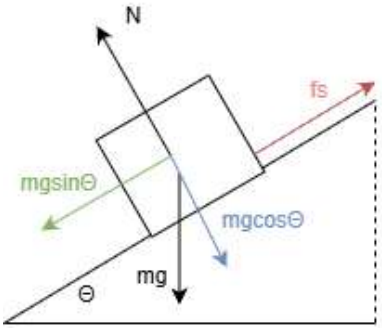
$f_{s30} = 0,091\ \text{N}$

$f_{s40} = 0,0882\ \text{N}$

$f_{ssebs} = 0,0152\ \text{N}$







$N_{30} := m_{shore} \cdot g \cdot \cos\left(\theta_{shore30}\right)$

$N_{40} := m_{shore} \cdot g \cdot \cos\left(\theta_{shore40}\right)$

$N_{sebs} := m_{sebs} \cdot g \cdot \cos\left(\theta_{sebs}\right)$

$b := \begin{bmatrix} \mu_{s30} \\ \mu_{ssebs} \\ \mu_{s40} \end{bmatrix}$

Maximum static friction

$\mu_{s30} := \frac{f_{s30}}{N_{30}}$ $\mu_{s30} = 2,4751$

$\mu_{s40} := \frac{f_{s40}}{N_{40}}$ $\mu_{s40} = 2,0503$

$\mu_{ssebs} := \frac{f_{ssebs}}{N_{sebs}}$ $\mu_{ssebs} = 0,2679$

$\mu_{preferred} := \max\left(b\right)$

$\mu_{preferred} = 2,4751$

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